

National College of Computer Studies

Paknajol, Kathmandu



Lab Report on Statistics

Assignment No: 3

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Bsc. CSIT (II Sem)

Submitted To:

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Practical No.: 3

The final grades in mathematics of 80 students are recorded in the accompanying table.

68	84	75	82	68	90	62	88	76	93	73	79
88	73	60	93	71	59	85	75	61	65	75	87
74	62	95	78	63	72	66	78	82	75	94	77
69	74	68	60	96	78	89	61	75	95	60	79
83	71	79	62	67	97	78	85	76	65	71	75
65	80	73	57	88	78	62	76	53	74	86	67
73	81	72	63	76	75	85	77				

Enter this data in Excel and generate following reports:

Questions:

- a) Construct a grouped frequency distribution by taking suitable class.

Step 1: Type your data into a worksheet. Make sure you put your data into columns. Use column headers. Type the data into cells A2 to A81 with a label for ease on A1 cell.

Step 2: To create a grouped frequency distribution table, we need to first determine the no. of classes and their size. For this, we calculate minimum, maximum and range of data.

For minimum:

For the calculation of the minimum data, we have used the predefined formula in excel. The raw data column is selected, and the corresponding formula is used,

$$= \text{MIN}(A2:A81)$$

$$= 53$$

For maximum:

For the calculation of the maximum data, we have used the predefined formula in excel. The raw data column is selected, and the corresponding formula is used,

$$= \text{MAX}(A2:A81)$$

$$= 97$$

For range:

For the calculation of the range of data, we have used the basic formula in excel.

$$= D3-D2$$

i.e., maximum - minimum

$$= 44$$

Step 3: For the calculation of the no. of classes, we have used the basic formula in excel.

$$\begin{aligned} &= 1+3.22*\text{LOG10}(80) && \text{i.e., } 1+3.22*\text{LOG10}(\text{total no. of data}) \\ &= 7.127 \sim 8 \end{aligned}$$

Step 4: For the calculation of class width, we have used the basic formula in excel.

$$\begin{aligned} &= D4/D5 && \text{i.e., range / no. of classes} \\ &= 6.172 \sim 6 \end{aligned}$$

Step 5: For creating the grouped frequency for the data, we, manually type the lower and upper limits of the 8 classes from 52-58 to 94-100. Then we calculate the bin limit for the class which is one less than the upper limit of the corresponding class and use the FREQUENCY formulae. The raw data column and the bin limit data is selected, and the corresponding formulae is used,

$$=\text{FREQUENCY}(\$A\$2:\$A\$81,\$E\$10:\$E\$17)$$

- b) Construct relative and percentage frequency distribution

For relative frequency distribution:

The proportion for the individual data is calculated by dividing the frequency for a particular data with the total sum of the frequency. The exact formula for a data is,

$$= F10/\$F\$18 \quad \text{i.e., frequency/sum of all frequencies}$$

For percentage frequency distribution:

For finding the proportion percentage we know multiply the obtained data in proportion cell by 100 to find the values on percentage. The exact formula for a data is,

$$= G10*100 \quad \text{i.e., proportion multiplied by 100}$$

- c) Draw a histogram for the grouped frequency distribution.

For creating a histogram, we create a grouped frequency distribution consisting of classes and their corresponding frequency. We select the grouped frequency distribution table and go to INSERT tab on the top ribbon. Find and click on the “bar chart” option in the “chart” section. Now you must click on the “Clustered Column” under the “2-D column” chart type there. After the chart has been created, go to the CHART DESIGN tab on the top ribbon and under “Chart Layout” section, click on “Quick Layout” and choose “Layout 8”.

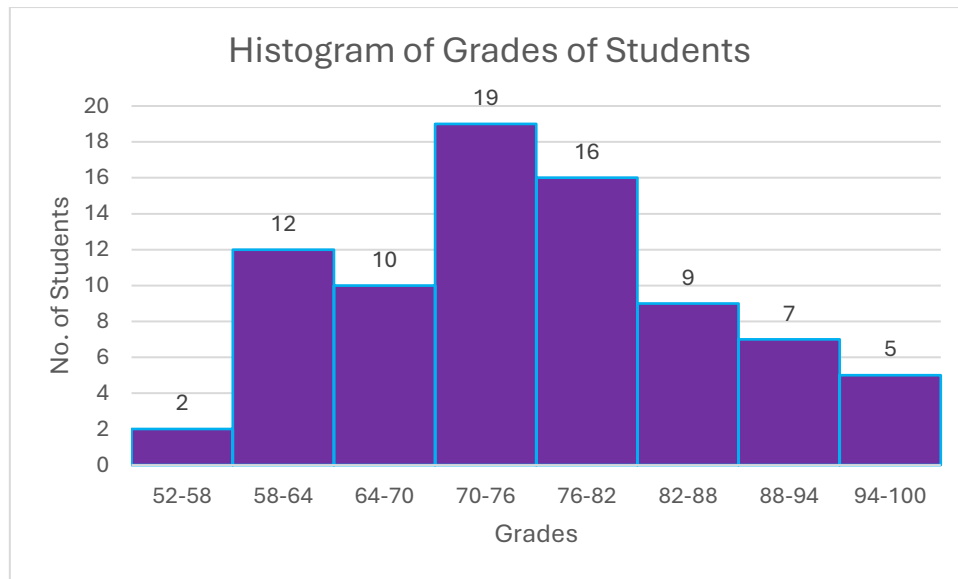


Fig 1: Histogram of Grades of Students

Interpretation:

From the above histogram of grades of students, more no. students, 19 & 16 have scored 70-76 and 76-82 grades, respectively. This means, on average, a student scores between 70-82. On the contrary, only 2 students scored in the range lowest 52-58. There is a sharp increase in the no. of students (12) scoring 58-64. 10 students scored 64-70, 9 of them scored 82-88, 7 scored 88-94 and only 5 of them scored in the highest range 94-100.

d) Find the Karl Pearson's coefficient of skewness (S_k) based on moment.

For the calculation of Karl Pearson's coefficient of skewness (S_k) based on moment, we need to find the mean, 1st, 2nd, 3rd and 4th order central moments. We have used the basic formulae for each of them. The raw data column is selected, and the corresponding formulae are used.

$$\text{Mean } (\bar{X}) = \text{AVERAGE}(A2:A81)$$

$$= 75.25$$

$$(X - \bar{X}) = A2 - \$D\$30$$

$$(X - \bar{X})^2 = C33 * C33$$

$$(X - \bar{X})^3 = D33 * C33$$

$$(X - \bar{X})^4 = E33 * C33$$

$$2^{\text{nd}} \text{ Order Central Moment } (m_2) = D113/80$$

$$= 106.2875$$

$$\text{i.e., } \frac{\sum (X - \bar{X})^2}{n}$$

$$3^{\text{rd}} \text{ Order Central Moment } (m_3) = E113/80 \quad \text{i.e., } \frac{\sum(X-\bar{X})^3}{n}$$

$$= 184.06875$$

$$4^{\text{th}} \text{ Order Central Moment } (m_4) = F113/80 \quad \text{i.e., } \frac{\sum(X-\bar{X})^4}{n}$$

$$= 27165.31953$$

For the calculation of the Karl Pearson's coefficient of skewness, we have used the basic formula in excel.

$$= D116/D115^{1.5} \quad \text{i.e., } \frac{m_3}{m_2^{1.5}}$$

$$= 0.17$$

Here,

D116 = 3rd Order Central Moment (m_3)

D115 = 2nd Order Central Moment (m_2)

e) Find the coefficient of kurtosis (k) based on moment

For the calculation of the Karl Pearson's coefficient of skewness, we have used the basic formula in excel.

$$= (D117/D115^2)-3 \quad \text{i.e., } \frac{m_4}{m_2^2} - 3$$

$$= -0.60$$

Here,

D117 = 4th Order Central Moment (m_4)

D115 = 2nd Order Central Moment (m_2)

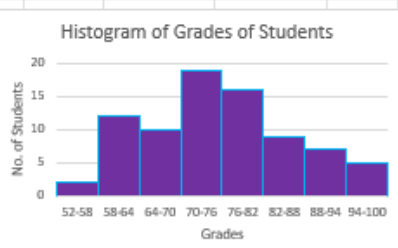
f) Draw as much as information about shape and size of the distribution

The following conclusions can be drawn about the shape and size of the distribution:

- i. Since the coefficient of skewness is 0.16, the distribution is slightly positively skewed i.e., almost symmetrical.
- ii. Since coefficient of kurtosis is -0.60, the distribution is platykurtic (low kurtosis)

Worksheet:

Y22														
1	Grades													
2	68	Minimum	53											
3	88	Maximum	97											
4	74	Range	44											
5	69	No. of Classes(k)	7.127949758											
6	83	Class Width(h)	6.172883016											
7	65		6 approx											
8	73													
9	84	Lower Limit	Upper Limit	Bin Limit	Frequency	Rel. Frequency	Per. Frequency							
10	73	52	58	57	2	0.025	2.5							
11	62	58	64	63	12	0.15	15							
12	74	64	70	69	10	0.125	12.5							
13	71	70	76	75	19	0.2375	23.75							
14	80	76	82	81	16	0.2	20							
15	81	82	88	87	9	0.1125	11.25							
16	75	88	94	93	7	0.0875	8.75							
17	60	94	100	99	5	0.0625	6.25							
18	95	Total			80	1	100							
19	68													
20	79	Grades	No. of Students											
21	73	52-58	2											
22	72	58-64	12											
23	82	64-70	10											
24	93	70-76	19											
25	78	76-82	16											
26	60	82-88	9											
27	62	88-94	7											
28	57	94-100	5											
29	63													
30	68	Mean	75.25											
31	71													
32	63	X-Mean	(X-Mean) ²	(X-Mean)	(X-Mean) ⁴									
33	96	-7.25	52.5625	-381.08	2762.816									
34	67	12.75	162.5625	2072.67	26426.57									
35	88	-1.25	1.5625	-1.9531	2.441406									
36	76	-6.25	39.0625	-244.14	1525.879									
37	90	7.75	60.0625	465.484	3607.504									
38	59	-10.25	105.0625	-1076.9	11038.13									
39	72	-2.25	5.0625	-11.391	25.62891									
40	78	8.75	76.5625	669.922	5861.816									
41	97	-2.25	5.0625	-11.391	25.62891									
42	78	-13.25	175.5625	-2326.2	30822.19									
43	75	-1.25	1.5625	-1.9531	2.441406									
44	62	-4.25	18.0625	-76.766	326.2539									
45	85	4.75	22.5625	107.172	509.0664									
46	66	5.75	33.0625	190.109	1093.129									
47	89	-0.25	0.0625	-0.0156	0.003906									
48	78	-15.25	232.5625	-3546.6	54085.32									
49	62	19.75	390.0625	7703.73	152148.8									
50	85	-7.25	52.5625	-381.08	2762.816									
51	88	3.75	14.0625	52.7344	197.7539									
52	75	-2.25	5.0625	-11.391	25.62891									
53	78	-3.25	10.5625	-34.328	111.5664									
54	61	6.75	45.5625	307.547	2075.941									
55	85	17.75	315.0625	5592.36	99264.38									
56	76	2.75	7.5625	20.7969	57.19141									
57	77	-15.25	232.5625	-3546.6	54085.32									
58	76	-13.25	175.5625	-2326.2	30822.19									
59	61	-18.25	333.0625	-6078.4	110930.6									
60	82	-12.25	150.0625	-1838.3	22518.75									
61	75	-7.25	52.5625	-381.08	2762.816									
62	76	-4.25	18.0625	-76.766	326.2539									
63	53	-12.25	150.0625	-1838.3	22518.75									
64	93	20.75	430.5625	8934.17	185384.1									
65	65	-8.25	68.0625	-561.52	4632.504									
66	75	12.75	162.5625	2072.67	26426.57									
67	95	0.75	0.5625	0.42188	0.316406									
68	65	14.75	217.5625	3209.05	47333.44									
69	74	-16.25	264.0625	-4291	69729									
70	73	-3.25	10.5625	-34.328	111.5664									
71	75	2.75	7.5625	20.7969	57.19141									
72	94	21.75	473.0625	10289.1	223788.1									



73	60	2.75	7.5625	20.7969	57.19141														
74	71	-0.25	0.0625	-0.0156	0.003906														
75	86	-13.25	175.5625	-2326.2	30822.19														
76	79	9.75	95.0625	926.859	9036.879														
77	87	-9.25	85.5625	-791.45	7320.941														
78	77	13.75	189.0625	2599.61	35744.63														
79	79	2.75	7.5625	20.7969	57.19141														
80	75	-13.25	175.5625	-2326.2	30822.19														
81	67	9.75	95.0625	926.859	9036.879														
82		12.75	162.5625	2072.67	26426.57														
83		-0.25	0.0625	-0.0156	0.003906														
84		2.75	7.5625	20.7969	57.19141														
85		-14.25	203.0625	-2893.6	41234.38														
86		9.75	95.0625	926.859	9036.879														
87		0.75	0.5625	0.42188	0.316406														
88		1.75	3.0625	5.35938	9.378906														
89		0.75	0.5625	0.42188	0.316406														
90		-14.25	203.0625	-2893.6	41234.38														
91		6.75	45.5625	307.547	2075.941														
92		-0.25	0.0625	-0.0156	0.003906														
93		0.75	0.5625	0.42188	0.316406														
94		-22.25	495.0625	-11015	245086.9														
95		17.75	315.0625	5592.36	99264.38														
96		-10.25	105.0625	-1076.9	11038.13														
97		-0.25	0.0625	-0.0156	0.003906														
98		19.75	390.0625	7703.73	152148.8														
99		-10.25	105.0625	-1076.9	11038.13														
100		-1.25	1.5625	-1.9531	2.441406														
101		-2.25	5.0625	-11.391	25.62891														
102		-0.25	0.0625	-0.0156	0.003906														
103		18.75	351.5625	6591.8	123596.2														
104		-15.25	232.5625	-3546.6	54085.32														
105		-4.25	18.0625	-76.766	326.2539														
106		10.75	115.5625	1242.3	13354.69														
107		3.75	14.0625	52.7344	197.7539														
108		11.75	138.0625	1622.23	19061.25														
109		1.75	3.0625	5.35938	9.378906														
110		3.75	14.0625	52.7344	197.7539														
111		-0.25	0.0625	-0.0156	0.003906														
112		-8.25	68.0625	-561.52	4632.504														
113	Total	0	8503	14725.5	2173226														
114																			
115		m2	106.2875																
116		m3	184.06875																
117		m4	27165.31953																
118																			
119																			
120		Coeff. Of Skewn	0.17	0.17121															
121																			
122		Coeff. Of Kurtos	-0.60	-0.5554															

1. Since coefficient of skewness is 0.16, the distribution is slightly positively skewed i.e almost symmetrical

2. Since coefficient of kurtosis is -0.59, the distribution is platykurtic (low kurtosis)